

LLM-Based RAG WhatsApp Bot Blueprint

1. Core Idea:

Self-hosted LLM + RAG + Agentic tools for MSMEs on WhatsApp with minimal external dependency.

2. Architecture:

Admin UI (React) -> Backend (FastAPI) -> RAG Engine + Agent Layer + DB + Storage -> Local LLM -> WhatsApp API

3. Key Modules:

- RAG Engine (FAISS, embeddings, document chunking)
- Agent Layer (booking, orders, DB actions)
- Multi-tenant system (separate business configs)
- WhatsApp integration (webhooks)

4. Storage:

- Local storage + optional Google Drive sync

5. Admin Dashboard:

- Upload/manage documents
- Customize prompts
- Enable/disable features
- Analytics

6. Database:

PostgreSQL with tables for businesses, users, messages, bookings, orders

7. RAG Pipeline:

Upload -> Chunk -> Embed -> Store -> Retrieve -> LLM

8. Use Cases:

Retail, restaurants, coaching centers, clinics

9. Roadmap:

Phase 1: Basic RAG

Phase 2: Multi-tenant

Phase 3: Agent tools

Phase 4: Scaling + analytics

10. Notes:

- Keep tenant isolation strict
- Optimize for low-resource environments
- Use quantized LLMs for performance